Exercise 14 - Exam Preparation

Exam Problem - HS23 - 2 minutes

Problem: In the following three questions, three different systems Σ_i, $i = 1, 2, 3$, are given by their input-output maps, where $u(t) \in \mathbb{R}$ denotes the input signal and $y(t) \in \mathbb{R}$ the corresponding output signal.

Q1 **(0.5 Points)** Mark all correct statements.

$$\Sigma_1 : \quad y(t) = 3\cos(t)u(t) + \int_0^t u(\tau - 4)\mathrm{d}\tau.$$

The system Σ_1 is,

- [x] Linear
- [B] Time-invariant
- [C] Static
- [x] Causal

$3\cos(t)u(t)$ is linear → also integral is a linear operation
time-variant due to $\cos(t)$
output depends on past input due to the integral → not static but causal

Q2 **(0.5 Points)** Mark all correct statements.

$$\Sigma_2 : \quad y(t) = \cos(u(t)) \int_{-\infty}^{t-2} u(\tau)\mathrm{d}\tau.$$

The system Σ_2 is,

- [A] Linear
- [x] Time-invariant
- [C] Static
- [x] Causal

$u(t)$ inside the $\cos(\cdot)$ is not a linear operation
time invariant $y(t-\tau) = (\Sigma\tilde{u})(t)$
not static due to integral → depends on past inputs ⇒ causal

Q3 **(0.5 Points)** Mark all correct statements.

$$\Sigma_3 : \quad y(t) = \sqrt{\cos\left(\frac{3\pi}{2}\right)} u(3t + 1).$$

The system Σ_3 is,

- [x] Linear
- [B] Time-invariant
- [C] Static
- [D] Causal

linear because we have a "constant" that multiplies $u(t)$
time variant due to the 3 inside u
↳ $y(t+\tau) = \dots u(3t + 3\tau + 1)$ $\neq$
$\neq$
delay only inside input $\dots u(3t+1+\tau)$
not static and non causal because depends on the input at future time

For more detailed explanations check Exercise 2 or solutions of HS23 exam

Exam Problem - FS24 - 3 minutes

Problem: Consider the following model of a dynamic system,

$$\begin{aligned} m\dot{z}(t)z(t) &= \frac{1}{\alpha} + \ddot{z}(t)v(t), \\ -\alpha\dot{v}(t) &= \frac{1}{F(t)} + z(t), \end{aligned}$$

where m and α are known system parameters and, where $z(t)$, $v(t)$ and $F(t)$ are signals.

Q5 (0.5 Points) Write the system in nonlinear state-space form $\dot{x}(t) = f(x(t), u(t))$, where the state $x(t)$ of the system is chosen to be

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} z(t) \\ \dot{z}(t) \\ v(t) \end{bmatrix},$$

and where the input $u(t)$ is chosen as

$$u(t) = \frac{1}{F(t)}.$$

$$\ddot{z}(t) = \frac{1}{v(t)}\left(m\dot{z}(t)z(t) - \frac{1}{\alpha}\right) \qquad u(t) = \frac{1}{F(t)}$$

$$\dot{v}(t) = -\frac{1}{\alpha F(t)} - \frac{z(t)}{\alpha}$$

$$\dot{x}(t) = \begin{pmatrix} \dot{z}(t) \\ \ddot{z}(t) \\ \dot{v}(t) \end{pmatrix} = \begin{pmatrix} x_2(t) \\ \frac{1}{x_3(t)}\left(m x_2(t) x_1(t) - \frac{1}{\alpha}\right) \\ -\frac{1}{\alpha}u(t) - \frac{1}{\alpha}x_1(t) \end{pmatrix}$$

Exam Problem - HS22 - 8 minutes

Problem: Consider the interconnected system shown in Figure 4. You can assume that the input-output relation for each system Σ_i is given by $y_i = \Sigma_i \cdot u_i$, that is, each of the systems Σ_i represents a simple scalar gain.

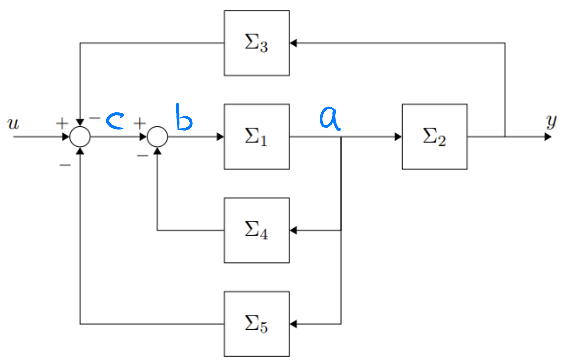


Figure 4: Interconnected system.

Q4 (1.5 Points) Derive the transfer function Σ from u to y for the interconnected system shown in Figure 4. **Simplify the result as much as possible.**

$$\Sigma = \frac{\Sigma_1\Sigma_2}{1 + \Sigma_1\Sigma_2\Sigma_3 + \Sigma_1\Sigma_4 + \Sigma_1\Sigma_5}$$

$$y = a\Sigma_2$$
$$a = b\Sigma_1$$
$$b = c - a\Sigma_4$$
$$c = u - a\Sigma_2\Sigma_3 - a\Sigma_5$$
$$b = u - a(\Sigma_2\Sigma_3 + \Sigma_4 + \Sigma_5)$$
$$a(1 + \Sigma_1\Sigma_2\Sigma_3 + \Sigma_1\Sigma_4 + \Sigma_1\Sigma_5) = u\Sigma_1$$
$$\Rightarrow y = \frac{\Sigma_1\Sigma_2}{1 + \Sigma_1\Sigma_2\Sigma_3 + \Sigma_1\Sigma_4 + \Sigma_1\Sigma_5}\, u$$

Exam Problem - HS23 - 5 minutes

Problem: Consider the nonlinear system with state $x(t) \in \mathbb{R}^2$, input $u(t) \in \mathbb{R}_{>0}$, and output $y(t) \in \mathbb{R}$ described by,

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 3e^{-x_1(t)+1} - 3x_1(t) \\ -\cos(x_2(t))\frac{1}{x_1(t)} - 2u^2(t)x_2^2(t) \end{bmatrix}, \\ y(t) &= x_1(t)x_2(t)u(t) + 1. \end{aligned}$$

Derive the (A, B, C, D) matrices of the state-space representation of the linearization of the above system around its unique equilibrium point $(x_{1,e} = 1,\ x_{2,e} = \pi,\ u_e = \frac{\sqrt{2}}{2\pi})$.

Q7 **(1 Points)** What is the A matrix of the linearization of the above system around the equilibrium $(x_{1,e},\ x_{2,e},\ u_e)$? The resulting A matrix must only contain numerical values.

Q8 **(0.5 Points)** What is the B matrix of the linearization of the above system around the equilibrium $(x_{1,e},\ x_{2,e},\ u_e)$? The resulting B matrix must only contain numerical values.

Q9 **(0.5 Points)** What is the C matrix of the linearization of the above system around the equilibrium $(x_{1,e},\ x_{2,e},\ u_e)$? The resulting C matrix must only contain numerical values.

Q10 **(0.25 Points)** What is the D matrix of the linearization of the above system around the equilibrium $(x_{1,e},\ x_{2,e},\ u_e)$? The resulting D matrix must only contain numerical values.

$$A = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{pmatrix}_{x_e,u_e} = \begin{pmatrix} -3e^{-x_{1e}+1} - 3 & 0 \\ \frac{\cos(x_{2e})}{x_{1e}^2} & -4u_e^2 x_{2e} \end{pmatrix}_{\underbrace{x_e,u_e}_{\text{we plug in everything}}} =$$

$$= \begin{pmatrix} -3 & 0 \\ -1 & -2/\pi \end{pmatrix}$$

$$B = \begin{pmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{pmatrix}_{x_e,u_e} = \begin{pmatrix} 0 \\ -4u_e x_e^2 \end{pmatrix}_{x_e,u_e} = \begin{pmatrix} 0 \\ -2\sqrt{2}\pi \end{pmatrix}$$

$$C = \begin{pmatrix} \frac{\partial h}{\partial x_1} & \frac{\partial h}{\partial x_2} \end{pmatrix}_{x_e,u_e} = \begin{pmatrix} x_{2e}u_e & x_{1e}u_e \end{pmatrix}_{x_e,u_e} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2\pi} \end{pmatrix}$$

$$D = \frac{\partial h}{\partial u}\Big|_{x_e,u_e} = x_{1e}x_{2e}\Big|_{x_e,u_e} = \pi$$

Exam Problem - FS24 - 4 minutes

Problem: Consider a linear time-invariant system with transfer function $G(s)$ that is given by a root-locus gain $k_{rl} = 1$ and by two poles and one zero as shown in Figure 2.

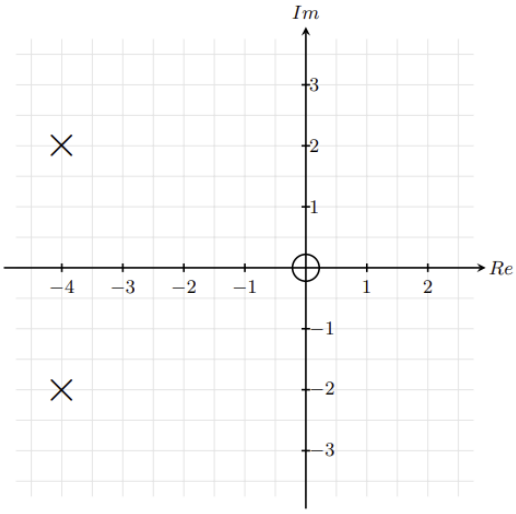


Figure 2: Poles (×) and zero (○) of the LTI system $G(s)$.

Further, the signal $u(t) = \sin(2t)$ is applied as the input to the system described by $G(s)$. It is known that the corresponding steady-state output signal is given by $y_{ss}(t) = M\sin(\omega t + \varphi)$.

Q16 **(0.5 Points)** Mark the correct answer.

What is the value of the frequency ω in the above expression of $y_{ss}(t)$?

- [A] $\omega = 0$ rad/s
- [C] $\omega = 2$ rad/s
- [B] $\omega = \sqrt{2}$ rad/s
- [D] $\omega = \frac{1}{2}$ rad/s

Output to a sinusoidal input is another sinusoidal of the same frequency but different magnitude and with a phase shift

Q17 **(1 Points)** Mark the correct answer.

What is the value of the magnitude M in the above expression of $y_{ss}(t)$?

- [A] $M = 5\sqrt{17}$
- [C] $M = \frac{1}{5\sqrt{17}}$
- [B] $M = 8\sqrt{2}$
- [D] $M = \frac{1}{8\sqrt{2}}$

Q18 **(1 Points)** Mark the correct answer.

What is the value of the phase φ in the above expression of $y_{ss}(t)$?

- [A] $\varphi = 45°$
- [C] $\varphi = -45°$
- [B] $\varphi = 0°$
- [D] $\varphi = 135°$

$$G(s) = \frac{s}{(s+4-2j)(s+4+2j)} \qquad \omega = 2\,\text{rad/s} \Rightarrow s = \omega j = 2j$$

$$M = |G(2j)| = \frac{|2j|}{|4|\cdot|4+4j|} = \frac{2}{4}\frac{1}{\sqrt{4^2+4^2}} = \frac{1}{2}\frac{1}{\sqrt{32}} = \frac{1}{8\sqrt{2}}$$

$$\varphi = \angle G(2j) = \angle(2j) - \angle(4) - \angle(4+4j) = \frac{\pi}{2} - 0 - \frac{\pi}{4} = \frac{\pi}{4}$$

Exam Problem - FS23 - 1 minute

Problem: Given in Figure 4 are the root locus plots of four different transfer functions for $k > 0$.

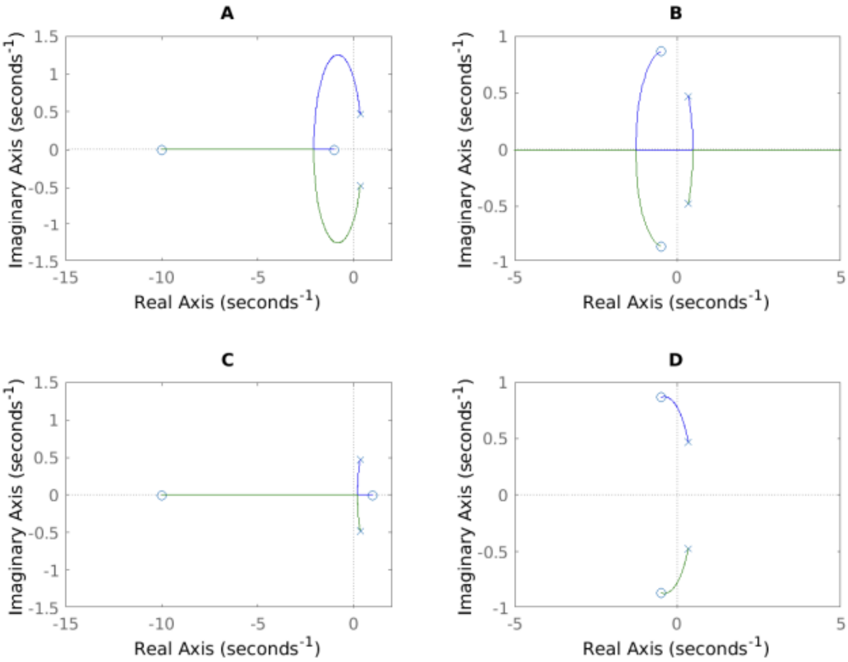


Figure 4: Root locus plots of four different systems. Poles and zeros are represented by $\times$ and o symbols respectively.

Q19 (1.5 Points) Assign each transfer function to the corresponding root locus plot.

Transfer Function	A	B	C	D
$L_1(s) = \dfrac{s^2+s+1}{3s^2-2s+1}$				X
$L_2(s) = -\dfrac{s^2+s+1}{3s^2-2s+1}$		X		
$L_3(s) = \dfrac{(s+10)(s-1)}{3s^2-2s+1}$			X	
$L_4(s) = \dfrac{(s+10)(s+1)}{3s^2-2s+1}$	X			

$L_1(s)$ and $L_2(s)$ have both complex poles and complex zeros

↳ but $L_2(s)$ has a negative constant ("inverse" root locus)

$L_1 \to D$ $L_2 \to B$ $L_4 \to A$

$L_3(s)$ has complex poles but zeros at $s=-10$ and $s=1$ $L_3 \to C$

Exam Problem - FS24 - 3 minutes

Problem: Consider the closed-loop system T shown in Figure 6, where L represents a linear time-invariant system and $k \in \mathbb{R}$. The root locus of L for $k > 0$ is shown in Figure 7.

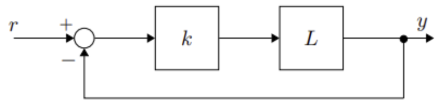


Figure 6: Closed-loop system T, open-loop system L and proportional gain k.

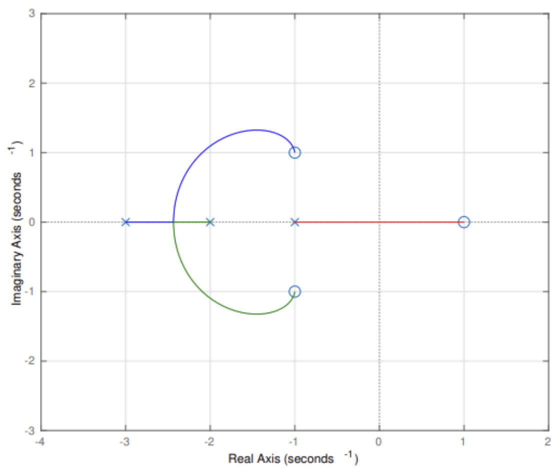


Figure 7: Root Locus of L. Poles are represented by $\times$ symbols, zeros by o symbols.

Q21 **(0.5 Points)** Mark the correct answer.
The open-loop system L is a minimum-phase system.

[A] True ■ False

Q22 **(0.5 Points)** Mark the correct answer.
The open-loop system L is asymptotically stable.

■ True [B] False

Q23 **(0.75 Points)** Mark the correct answer.
There exists a gain $k^\star$ such that for all k such that $0 < k \leq k^\star$ the closed-loop system T is asymptotically stable and all poles of the closed-loop system have zero imaginary part, i.e. are real numbers.

■ True [B] False

Q24 **(1 Points)** Mark the correct answer.
Does there exist a gain $k_{\text{crit}} > 0$ such that for all $k > k_{\text{crit}}$ the closed-loop system T is unstable? If so, what is value of k_{crit}?

[A] $k_{\text{crit}} = \frac{1}{\sqrt{2}}$ [C] $k_{\text{crit}} = \frac{1}{3}$

■ $k_{\text{crit}} = 3$ [D] There is no such k_{crit}.

$$|L(s)| = \frac{1}{|k|}$$

$$L(s) = \frac{(s-1)(s+1-j)(s+1+j)}{(s+1)(s+2)(s+3)}$$

$$|L(s=0)| = \frac{|-1||1-j||1+j|}{|1||2||3|} = \frac{1\sqrt{2}\sqrt{2}}{2\cdot 3} = \frac{1}{3} = \frac{1}{|k|} \quad k=3$$

the closed loop becomes unstable when $s=0$
↳ closed loop pole cross the im. axis

Exam Problem - HS22 - 2 minutes

Problem: Given in Figure 9 are the Bode plots of four different transfer functions.

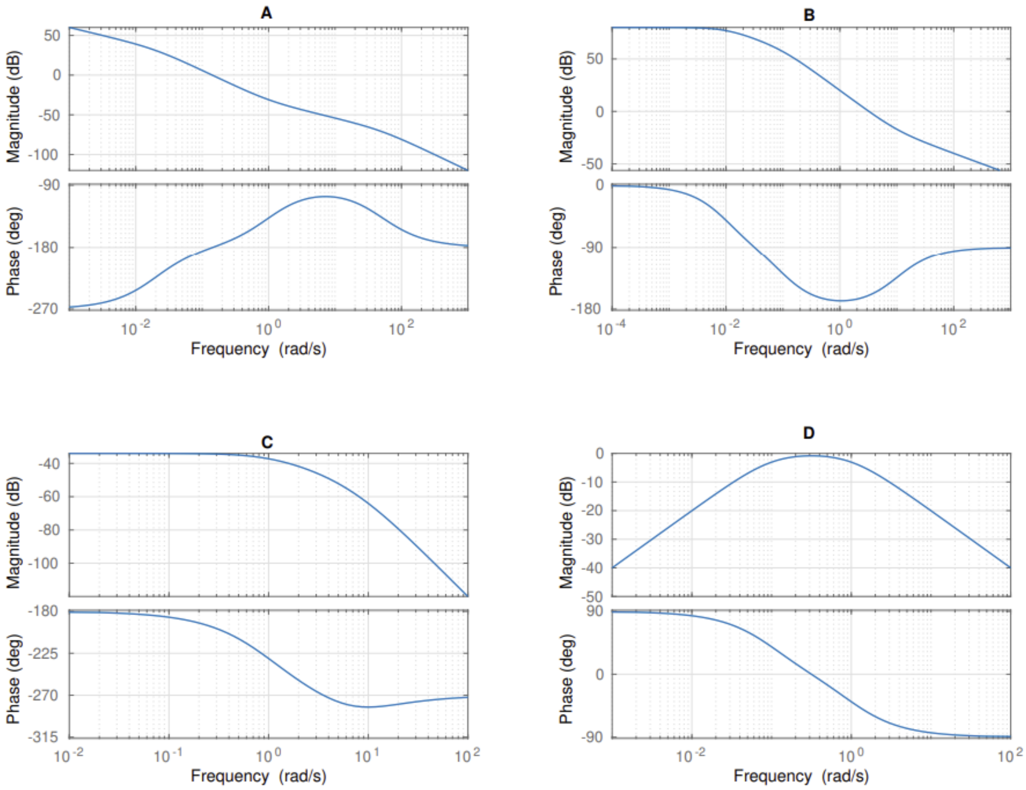


Figure 9: Bode plots of four different systems.

Q23 **(1.5 Points)** Assign each transfer function to the corresponding Bode plot.

Transfer Function	A	B	C	D
$L_1(s) = \frac{s+1}{s(s+50)(s-0.02)}$	X			
$L_2(s) = \frac{s}{(s+1)(s+0.01)}$				X
$L_3(s) = \frac{s+10}{(s+0.01)(s+0.1)}$		X		
$L_4(s) = \frac{1}{(s+1)(s-10)(s+5)}$			X	

A starts with decreasing magnitude → integrator ⇒ $A \to L_1$

D starts with increasing magnitude (also 90° phase): differentiator

⇒ $D \to L_2$

$L_4(s \to 0) = \frac{1}{-50} \simeq 30\,dB$ and $\pm 180°$ phase ⇒ $C \to L_4$

$B \to L_3$

Exam Problem - FS24 - 1 minute

Problem: Consider the transfer functions,

$$G_1(s) = \frac{s}{s^2+2s+0.1},\ G_2(s) = e^{-s}\frac{s}{s^2+2s+0.1},\ G_3(s) = \frac{s+10}{(s+1)(s+5)},\ G_4(s) = \frac{s+10}{(s+1)(s-5)}.$$

In Figure 8 the Bode plots of the aforementioned transfer functions are shown in a randomized order.

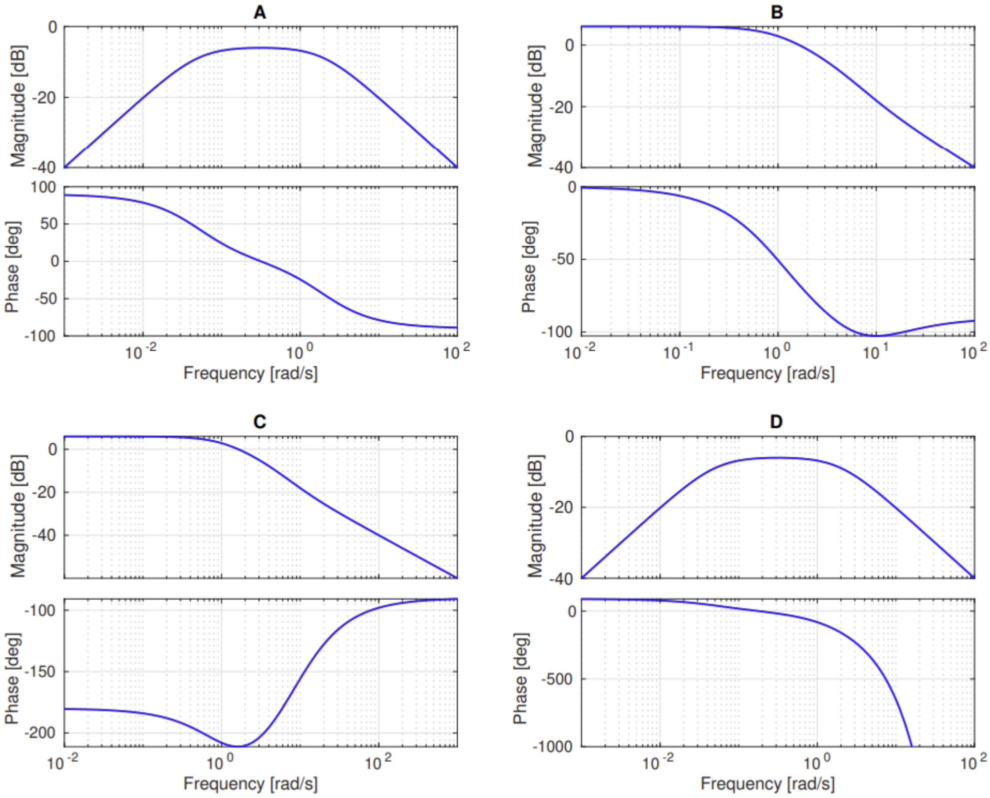


Figure 8: Bode plots of G_1, G_2, G_3 and G_4 in random order.

Q25 **(1 Points)** Mark the correct answer.
Mark the correct Bode plot and transfer function pairings?

- [A] (A, G_1), (B, G_3), (C, G_4), (D, G_2)
- [B] (A, G_2), (B, G_3), (C, G_4), (D, G_1)
- [C] (A, G_1), (B, G_4), (C, G_3), (D, G_2)
- [D] (A, G_3), (B, G_1), (C, G_2), (D, G_4)

G_2 has a time delay $\rightarrow$ we see its effect in the phase of D $\rightarrow$ phases decreases really fast

$G_2 \rightarrow D$

$G_4(s\to 0) = \frac{10}{1\cdot(-5)} = -2 \Rightarrow$ phase $\pm 180°$ magnitude >0

$\hookrightarrow$ must be C $\Rightarrow$ $G_4 \rightarrow C$

$G_1 \rightarrow A \Rightarrow$ differentiator $+20$dB/dec $\quad G_3 \rightarrow B$

Exam Problem - FS23 - 4 minutes

Problem: Consider the closed-loop system T in Figure 9, where L is a linear time-invariant system.

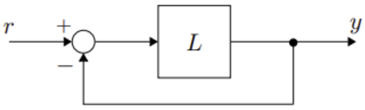


Figure 9: Closed-loop system T.

Assume that $L(s)$ is given by

$$L(s) = k\frac{s+a}{(s+b)(s+c)},$$

with $k, a, b, c \in \mathbb{R}$ unknown. Through two experiments you are able to derive the Nyquist plot of the system which is shown in Figure 10, and you are able to determine that the zero a is non-minimum phase, i.e. $a < 0$.

Remember

$Z = N + P$

unstable closed-loop poles

unstable open-loop poles

encirclements

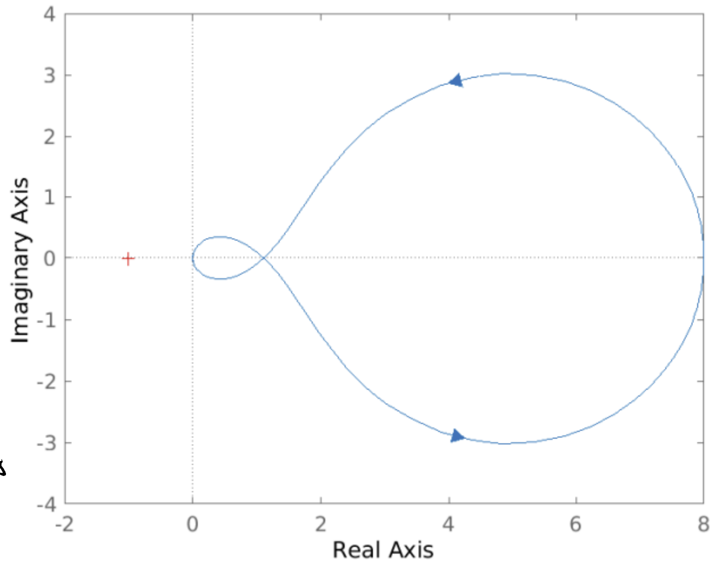


Figure 10: Nyquist plot of $L(s)$.

Q26 **(1.5 Points)** What is the number of unstable poles Z of the closed-loop system T?

$Z = 1$

More poles than zeros $\Rightarrow$ $L(s \to \infty) = 0$

This means that we start at the point $P = 8$

We have a non-minimum zero $\rightarrow$ $+20$dB/dec and $-90°$ phase

To close at $-90°$ with two poles we need to have an unstable pole and a stable pole ($-90°$ and $+90°$ phase $= 0° - 90°$ of zero $= -90°$)

Thus $Z = N + P = N + 1$ $\Rightarrow$ at -1 no encirclements $\Rightarrow$ $Z = 1$

Exam Problem - FS23 - 2 minutes

Problem: Consider the time responses of four different second order linear time-invariant systems described by four transfer functions $G_i(s)$, $i = 1, 2, 3, 4$, given in Figure 11. The bode plots corresponding to the aforementioned systems are shown in Figure 12.

G_3 and G_2 exhibit oscillations
↳ we only have that behavior in B and C → small peak
in C the peak is at higher frequency than B → C is "faster" than B → C → G_2 B → G_3
For A and D we see the behaviors of the poles at higher frequencies in D
↳ D stabilizes first (faster)
D → G_1 A → G_4

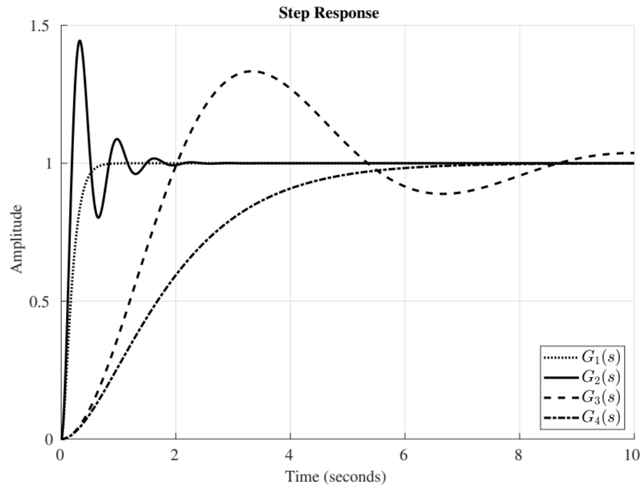
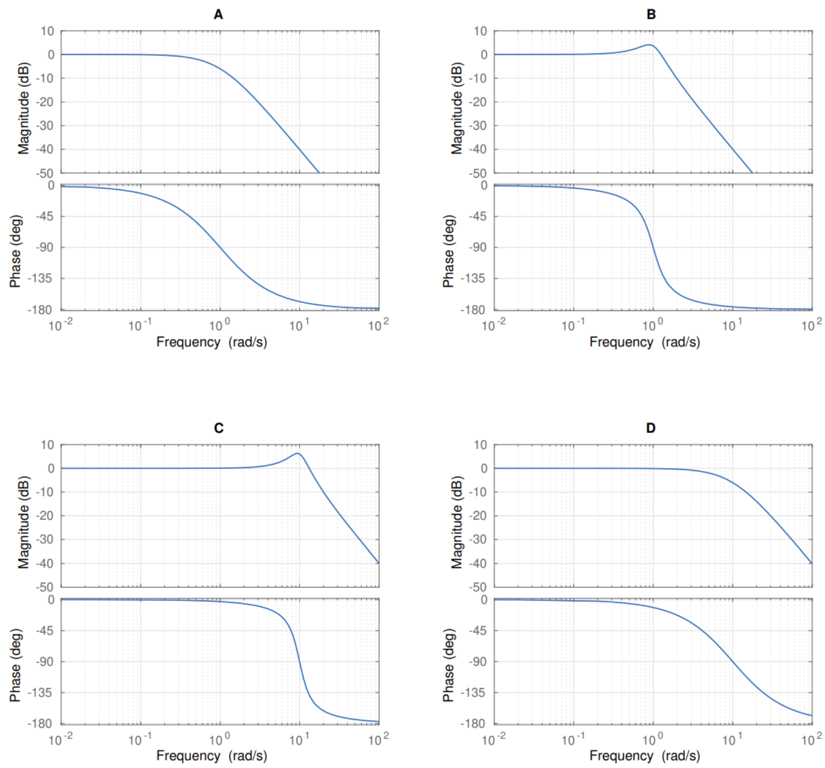


Figure 11: Step responses of second order linear-time invariant systems $G_i(s)$, $i = 1, 2, 3, 4$.

Q27 **(1.5 Points)** Assign each transfer function to the corresponding Bode plot.

Transfer function	A	B	C	D
$G_1(s)$				X
$G_2(s)$			X	
$G_3(s)$		X		
$G_4(s)$	X			



Exam Problem - HS23 - 3 minutes

Problem: Consider the transfer functions,

$$G_1(s) = \frac{20(s+5)}{(s+2)(s-50)},\quad G_2(s) = \frac{20(s-5)}{(s+2)(s+50)},\quad G_3(s) = \frac{20(s-5)(s+10)}{(s+2)(s-50)},\quad G_4(s) = \frac{e^{-2s}}{s}.$$

In Figure 16, the Nyquist plots of the aforementioned transfer functions are shown in a randomized order.

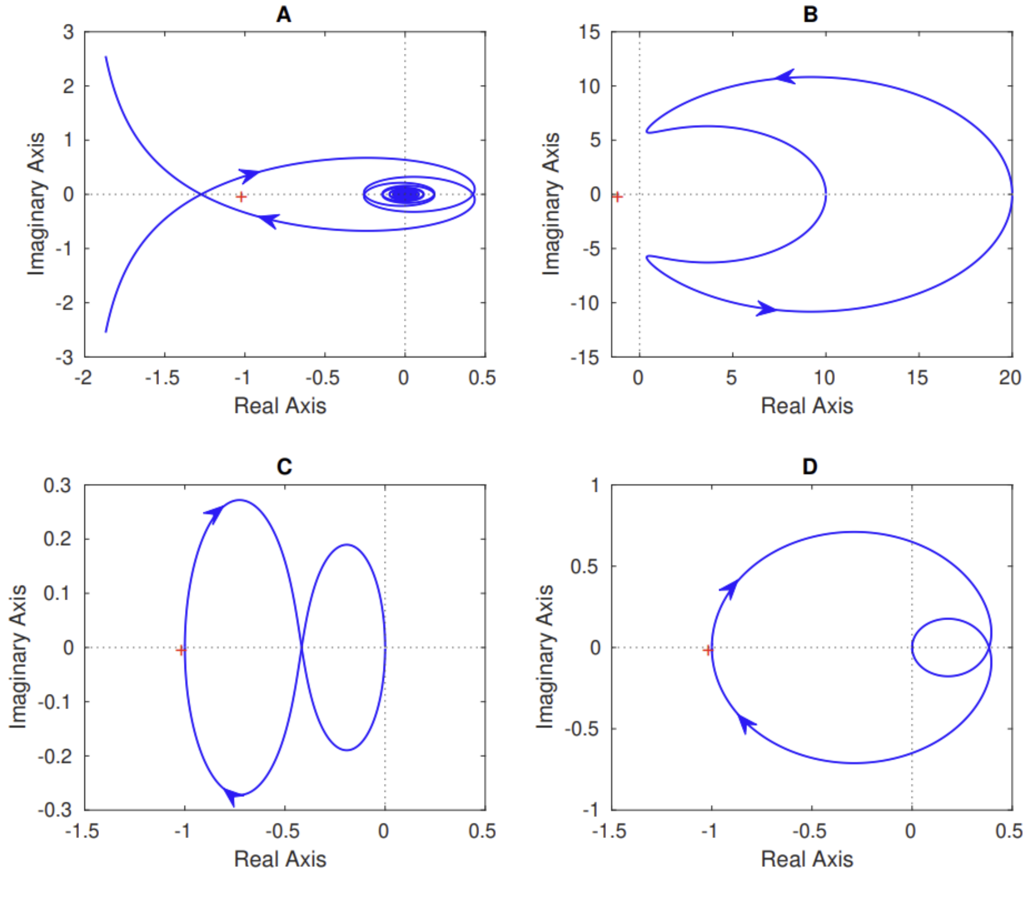


Figure 16: Nyquist plots of G_1, G_2, G_3 and G_4 in random order.

Q33 **(1 Points)** Mark the correct answer.
Mark the correct Nyquist plot and transfer function pairings?

- [A] (A, G_3), (B, G_2), (C, G_1), (D, G_4)
- [B] (A, G_4), (B, G_1), (C, G_3), (D, G_2)
- [C] (A, G_4), (B, G_3), (C, G_2), (D, G_1)
- [x] (A, G_4), (B, G_3), (C, G_1), (D, G_2)

We can note that the circling behavior that we see in A corresponds to a time delay $\Rightarrow$ therefore $G_4(s) \rightarrow A$

Then for $G_3(s \to 0) = \frac{20\cdot(-5)\cdot 10}{2\cdot(-50)} = 10 \Rightarrow$ only option is B

$\Rightarrow G_3(s) \rightarrow B$

For G_1 we start at -1 $G_1(s\to 0) = -1$

Then we encounter first a stable pole $(s=-2) \Rightarrow -90°$ phase

Then we encounter a min. phase zero $(s=-5) \Rightarrow +90°$ phase

$\Rightarrow$ We see this behavior in C : $G_1 \rightarrow C$ $G_2 \rightarrow D$

Exam Problem - FS24 - 10 minutes

Problem: Let P denote a linear time-invariant system that is to be controlled with controller C using a standard feedback architecture. The system is subject to measurement noise n, as shown in Figure 15.

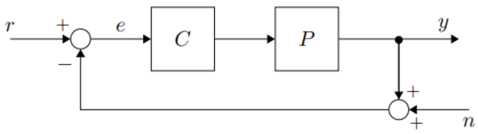


Figure 15: Standard feedback architecture with measurement noise. Plant P, controller C, measurement noise n.

Suppose that C is some proportional controller, and that the corresponding closed-loop system T has the following strictly proper rational transfer function:

$$T(s) = \frac{C(s)P(s)}{1+C(s)P(s)} = \frac{0.5}{s^2+0.6s+0.25}.$$

The closed-loop system, when subject to a unit step input, has the following control requirements:

1. A settling time of 10 s within $d = \frac{100}{e^2}\%$ of the steady-state.
2. A maximum overshoot $M_p = \frac{1}{e}$.
3. No steady state error.

Q33 **(1 Points)** Mark the correct answer.
Is the settling time requirement, i.e. requirement 1, for the closed-loop system T satisfied?

A Yes
B No
C Cannot be determined from the provided information

Q34 **(1 Points)** Mark the correct answer.
Is the overshoot requirement, i.e. requirement 2, for the closed-loop system T satisfied?

A Yes
B No
C Cannot be determined from the provided information

Q35 **(1 Points)** Mark the correct answer.
Neglecting control requirements 1 and 2, which of the following controller types is required to satisfy requirement 3? If there exist multiple controller types that could satisfy requirement 3, select the one so that the resulting closed-loop system is the least sensitive to measurement noise.

A P Controller, $C(s) = \kappa_P$.
B PI Controller, $C(s) = \kappa_P + \frac{\kappa_I}{s}$.
C PD Controller, $C(s) = \kappa_P + \kappa_D s$.
D PID Controller, $C(s) = \kappa_P + \kappa_D s + \frac{\kappa_I}{s}$.

also PID possible
↳ but derivative term increase noise sensitivity

$T_d = -\frac{1}{\sigma}\ln\left(\frac{d}{100}\right) \Rightarrow d = 100/e^2$

$\Rightarrow T_d = -\frac{1}{\sigma}\ln\left(\frac{1}{e^2}\right) = \frac{2}{\sigma}$

Recall that $\sigma = \zeta\omega_n$ and that $G(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

$\Rightarrow 0.6 = 2\zeta\omega_n \Rightarrow 0,3 = \zeta\omega_n = \sigma$

$\Rightarrow T_d = \frac{2}{0,3} \approx 6,67s < 10s$

$\zeta_r \approx 0,3$

$\zeta_r^2 = \frac{(\ln(M_p))^2}{\pi^2+(\ln(M_p))^2} = \frac{(\ln(\frac{1}{e}))^2}{\pi^2+(\ln(\frac{1}{e}))^2} = \frac{1}{1+\pi^2} \approx 0,092$

$\Rightarrow \omega_n = 0,5 \Rightarrow \zeta = 0,6 > \zeta_r = 0,3$ ✓

Exam Problem - HS23 - 10 minutes

Problem: Let P represent a linear time-invariant system that is to be controlled using a standard feedback architecture with controller C as shown in Figure 20. The closed-loop system is referred to as T. The Bode plot of the transfer function of P is given in Figure 21.

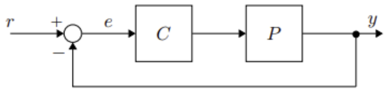


Figure 20: Closed-loop system T with plant P and controller C.

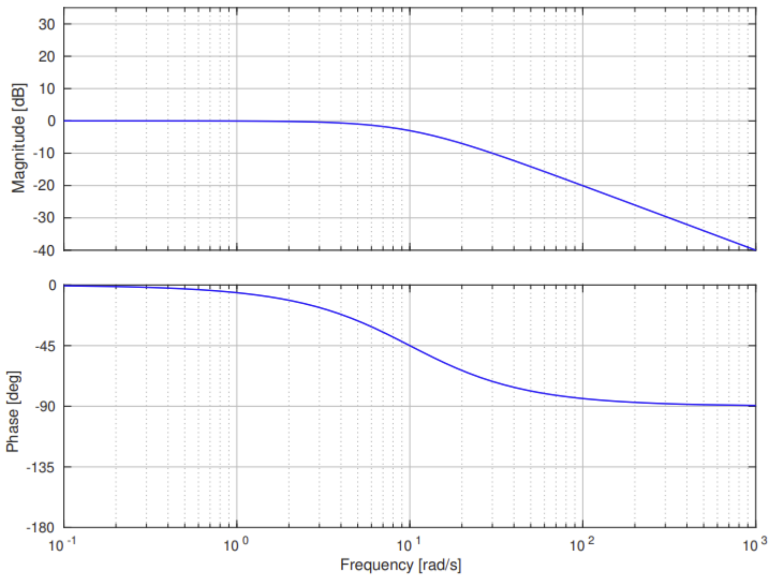


Figure 21: Bode plot of P.

The desired control objectives are,

- no steady-state error to step references,
- tracking error to sinusoidal inputs of frequencies up to 1 rad/s should not exceed 10% in magnitude,
- (noise) frequencies higher than 50 rad/s should be suppressed at least by a factor 10,
- desired phase margin of at least 45°, and,
- a desired closed-loop bandwidth of approximately 10 rad/s.

Q37 **(1.5 Points)** Mark the correct answer.
Which of the following controller designs $C(s)$ best meets the above control requirements?

- A $C(s) = \frac{0.1}{s}$
- B $C(s) = \frac{s+1}{s+10}$
- C $C(s) = \frac{10}{s}$
- D $C(s) = \frac{s+10}{s+1}$
- E None of the given choices closely meets the desired control objectives.

$P(s) \approx \frac{10}{s+10}$ $\Rightarrow$ there is a stable pole at $s = -10$
$-90°$ phase and -20dB/dec

• steady state error $e_{ss} = \frac{1}{1+L(0)} = \frac{1}{2}$ $\Rightarrow$ we must have an integrator to have zero steady-state error

• $|S(j\omega)| < 0{,}1$ when $\omega < 1$ rad/s

$\Rightarrow \frac{1}{|1+L(j\omega)|} < 0{,}1 \Leftrightarrow 10 < |1+L(j\omega)| \approx |L(j\omega)|$ now we have $|L(j\omega)| \approx 0$

only possible with option C and D (but we need integrator $\rightarrow$ C)

• $|T(50j)| < \frac{1}{10} |P(50j)|$ $\Rightarrow$ only option C

$\Rightarrow$ Then check if C satisfies the other two requirements

$|L(j\omega)| = \left|\frac{10}{j\omega_{gc}}\right| \left|\frac{10}{j\omega_{gc}+10}\right| = 1 \Rightarrow \omega_{gc} \approx 10$ rad/s $\quad \angle L(j\omega_{gc}) = -90° - 45° = -135°$

Exam Problem - HS22 - 4 minutes

Problem: Consider a lag compensator

$$C_{\text{lag}}(s) = \frac{a}{b} \cdot \frac{s/a+1}{s/b+1}, \quad 0 < b < a, \tag{2}$$

used in a standard feedback architecture to control a given plant P.

Q39 **(0.5 Points)** Mark the correct answer for each statement.

Check definitions and characteristics of lag compensator in Exercise 11

Statement	True	False
A lag compensator is frequently used to improve command tracking/disturbance rejection.	X	
A lag compensator cannot cause the system to become unstable.		X
The slope of the magnitude at frequencies between a and b is approximately -20 dB/decade.	X	

because it reduces phase margin

Q40 **(1.5 Points)** A lag compensator is to be used in order to improve the steady-state error to a step response to less than 2%. In order to achieve this, the low-frequency magnitude of the open-loop transfer function needs to be increased by a factor of 5. The current control design already features a satisfying phase margin and a satisfying open-loop gain crossover frequency ω_{gc} currently at 100 rad/s.

Which choice of the parameters a and b below satisfy aforementioned steady-state error requirement and have least impact on the crossover frequency and phase margin? Mark the correct answer.

- [ ] $a = 15$, $b = 75$
- [ ] $a = 1$, $b = 5$
- [x] $a = 1$, $b = 0.2$
- [ ] $a = 75$, $b = 15$

- Steady state error $< \frac{2}{100}$
- Low frequency magnitude increased by a factor of 5
- ω_{gc} at 100 rad/s

$b = 75$ or $b = 5 > a$
↳ not a lag compensator

$\frac{a}{b} = 5 \Rightarrow b = 0{,}2$ or $b = 15$

$\Rightarrow$ With $b = 15$: $a = 75 \Rightarrow$ very close to 100 rad/s
↳ more impact on the phase margin

$\Rightarrow$ correct answer is $a = 1 \quad b = 0.2$

Exam Problem - HS23 - 1 minute

<u>Problem:</u> Address the question below.

Q42 **(1 Points)** Mark all correct statements.

- [A] The circle criterion is a necessary and sufficient condition for absolute stability.
- [B] Describing functions can be seen as approximate transfer functions that represent the transfer function from sinusoidal inputs of amplitude A and frequency ω to the first harmonic of the output of the corresponding non-linearity.
- [C] Limit cycles are typically observed in linear systems.
- [D] The describing function of a static non-linearity is an amplitude dependent gain $N(A)$.

Check definitions and characteristics in Exercise 13